

BAYESIAN TOLERANCE INTERVALS FOR THE BALANCED ONE-WAY RANDOM EFFECTS MODEL WITH t -DISTRIBUTED MEASUREMENT ERRORS

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Summary: In this paper we consider the estimation of variance components in the one-way random effects model and consider the assessment of a manufacturing process's performance when a sample of parts produced by the process is measured with error. This estimation of the variance components when a non-standard variance components model is needed will be considered from a Bayesian point of view. Approximate posterior distributions will be obtained which can be used in the balanced case, but which can with relative ease, be converted also for use in the unbalanced case. In addition to point estimates, 95% credibility intervals as well as tolerance intervals will be provided.

1. Introduction

The practice of quality engineering in the manufacturing environment is changing rapidly, with many companies facing higher demands with the introduction of new systems and new products. There is furthermore an increased pressure for quality engineers as well as other manufacturing activities to support the economic objectives and profitability of the firm. More tools are needed by quality engineers to cope with these changes and to meet the intense international competition (Black Nembhard & Valverde – Ventura, 2003).

Manufacturers are thus frequently required to verify that products meet certain specifications (Hahn, 1982). A standard approach to the problem is to compare for example measurements from a sample of parts, to a certain specification. Inferences can then be made from these results about the entire population of parts (Wilson, Hamada & Xu, 2004). Sometimes, however, the situation arises when, for example, it may seem that specifications are not being met, when in fact they are. This usually occurs when the available data is subject to measurement error (Hahn, 1982). Therefore one must account for the measurement system being used to characterize production performance based on, for example, the sample of parts (Wilson, Hamada & Xu, 2004).

Variance component models are needed to assess the manufacturing process's performance when the measurement error variance depends on the true characteristics of, for example the parts being measured (Wilson, Hamada & Xu, 2004). These variance component models are frequently used in quality control. Variances are usually estimated for balanced data using the minimum variance unbiased estimators (MVUEs). These MVUEs are based on the sums of squares appearing in the analysis of variance (ANOVA) table.

MVUEs, however, do not exist for unbalanced data, since the sums of squares from the ANOVA table are not sufficient statistics (Chaloner, 1987). Searle (1979) presented various other estimators, for example, maximum likelihood estimators (MLEs), restricted maximum likelihood estimators (REMLs), minimum norm quadratic unbiased estimators (MINQUEs), minimum variance quadratic unbiased estimators (MIVQUEs), as well as several variations of these approaches. The use of analysis of variance (ANOVA) estimators that are obtained by equating mean squares to their expected values, is also another common approach (Chaloner, 1987). The Bayesian methodology can also be used to assess a manufacturing process's performance and thus provides a flexible alternative to process assessment through variance component and interval estimation.

To illustrate how a process's performance can be assessed when a sample of parts produced by a process is measured with error, consider the following example from a new manufacturing process. The data in Table 1 was obtained from a new manufacturing process and represents measurements of iron concentration in parts per million (ppm) (Wilson, Hamada & Xu, 2004). Each row of Table 1 represents two measurements of the same part determined by emission spectroscopy. According to Wilson, Hamada and Xu (2004) a part is considered to be acceptable if it has under 225 ppm of iron. The engineers involved in the process are interested in understanding production characteristics while the chemists, who measure the parts, are interested in understanding the measurement system (Wilson, Hamada & Xu, 2004).

Wilson, Hamada and Xu (2004) suggest that a one-way model be used which is mathematically given by

$$y_{ij} = x_i + \epsilon_{ij} \quad (1)$$

Table 1 Measurements of iron in parts per million (ppm) determined by emission spectroscopy. (A part with under 225 ppm of iron is acceptable.)

Part	Measurement 1	Measurement 2
1	206	258
2	181	197
3	185	162
4	195	195
5	170.5	143.8
6	193.5	224.8
7	244.8	217
8	191.5	196.8
9	209.3	189.5
10	134.5	143.8
11	223.8	198.5
12	103	129.3
13	99.7	201.8
14	137.5	119.8
15	144.5	130
16	159	166.5
17	140.5	138
18	207	230
19	195.5	190.5
20	142.3	163.8
21	74.3	86.5
22	439.5	211.5
23	130.3	114
24	99.7	201.8

where y_{ij} is the j^{th} measurement on the i^{th} part, x_i is the true value for the i^{th} part and ϵ_{ij} is the measurement error for the j^{th} measurement on the i^{th} part. We are really interested in x_i and ϵ_{ij} only, but what are observed are the y_{ij} 's. A variety of issues can be addressed if it is possible to estimate

the distributions of x_i and ϵ_{ij} . These include many of the common ways of characterizing production performance.

The novel features of the above example arise because the measurement system cannot be characterized in the traditional way, i.e. as following a $N(0, \sigma_\epsilon^2)$ distribution for ϵ_{ij} . Rather, the measurement system may not be normally distributed and may have, for example, a multiplicative (or different) structure for the variance (Wilson et al. 2004). The remainder of this paper will be dedicated to the characterization of the measurement system in Table 1 using a t -distribution.

Since the Bayesian approach provides an alternative approach to the assessment of measurement systems, and also based on the above considerations, it was decided to use the Bayesian approach for the estimation of the variance components. The next section presents some background information on the classical approach as well as the Bayesian approach to variance component estimation. This is followed by the computational methods for implementing the Bayesian approach. The Bayesian method is then applied to the data in Table 1 and to conclude we will also provide Bayesian tolerance intervals in the case where the residuals follow a t -distribution.

2. The variance component model

For our example, it is necessary to employ a statistical model which adequately measures multiple sources of variability, since there is variability between parts and also within parts (see Table 1). It was therefore decided to use the statistical model given by equation 1 which is known as a variance components model.

Most authors assume that the production or part measurements, x_i ($i = 1 \dots I$), follow independent $N(0, \sigma_p^2)$ distributions, and that the ϵ_{ij} ($i = 1 \dots I$

and $j = 1 \dots J$) follow independent $N(0, \sigma_\epsilon^2)$ distributions. The model has been considered in a variety of contexts. Hahn (1982) estimated the proportion of parts that meet specification under the assumption that the residual variance σ_ϵ^2 was known. Jaech (1984) also considered the case where the error variance σ_ϵ^2 was known. Tolerance intervals for the proportion of parts meeting specification were then estimated. Mee (1984b) also estimated tolerance intervals, but considered the cases where σ_ϵ^2 was known, the ratio of $\sigma_\epsilon^2/\sigma_p^2$ was known and the case where the ratio of the variances was estimated through repeated measurements. More complicated models for the parts distributions (e.g. random effects, random coefficients and mixed effects) were considered by Wang and Iyer (1994). They also calculated tolerance intervals. Note also that these authors all used a frequentist perspective to approach the variance components problem.

The Bayesian approach provides an alternative approach to the problem of variance component estimation for the one-way random effects model given in equation 1. The Bayesian approach to variance component estimation has been studied extensively for balanced data where $\epsilon_{ij} \sim N(0, \sigma_\epsilon^2)$ (for $i = 1 \dots I$ and $j = 1 \dots J$). The Bayesian approach for balanced data was successfully implemented by Tiao and Tan (1965) and Box and Tiao (1973). These authors obtained some approximations for posterior densities of variance components and also derived some closed-form estimators for these components of variance. Hill (1977) developed exact and approximate Bayesian solutions for inference about variance components. The sampling properties of Bayesian and other estimators were investigated by Klotz, Milton and Zacks (1969). Skene (1983) provided computational methods for obtaining marginal posterior densities for variance components in the case of crossed and nested models.

There is, however, little difference between the balanced and unbalanced case in a Bayesian context (Chaloner, 1987). Hill (1965) obtained exact and approximate posterior distributions for variance components in the balanced and unbalanced cases. Hill (1965) also discussed and provided the necessary integrations. Chaloner (1987) provided a basic Bayesian approach to the unbalanced one-way variance components model using a non-informative prior distribution that is uniform on the intraclass correlation. Chaloner (1987) also illustrated that although the ANOVA estimators can easily be calculated by hand, the Bayesian estimators are generally much more efficient than these ANOVA estimators. In addition to these, Chaloner (1987) also provided a simulation study for the estimation of the ratio of the variance components and investigated the sampling properties of the highest posterior density regions for this ratio. It was also pointed out by the author that although these highest posterior density regions might not have the coverage probabilities of confidence intervals, they do lead to sensible interval estimates that are never empty. Analytical results for the estimation of a production process were provided by Hahn and Raghunathan (1988).

The Bayesian approach to variance component estimation was revisited after the development of Markov Chain Monte Carlo methods by authors such as Gelfand, Hills & Racine-Poon (1990), Sun, Hsu, Gutman & Leonard (1996), Wolfinger (1998), Wolfinger and Kass (2000) and more recently Wilson et al. (2004). Sun et al. (1996) demonstrated that with the advent of powerful techniques such as importance sampling, Markov chain iterations and modern usage of Laplacian approximations, it became possible to provide detailed finite sample inference for many variance component models. Also, since the computations could be handled more efficiently, importance sampling, Laplacian approximations and the Gibbs sampler permitted the consideration

of models and prior assumptions of high complexity. Sun et al. (1996) also showed that the Bayesian estimates of the first stage parameters have excellent frequentist properties when non-informative priors are assumed for variance components. The authors also mentioned that if prior knowledge was available, these estimates could further be improved in a subjective sense, through the use of inverted chi-square distributions for the variance components. Interval estimates and posterior probabilities were also readily available.

Wolfinger (1998) used Markov chain Monte Carlo simulation techniques to generate a random sample from the joint posterior distribution of the mean and the variance parameters to construct a sample from other relevant posterior distributions. Wolfinger (1998) also presented a simulation-based approach for determining Bayesian tolerance intervals in variance component models and illustrated that different kinds of tolerance intervals could be determined in a straightforward way. It was also pointed out that this methodology could easily be tailored to particular applications. Wolfinger and Kass (2000) indicated that although Gibbs sampling is easy to implement for balanced data using conjugate priors, simulating from full conditional posterior densities can become difficult for the analysis of unbalanced data with possibly non-conjugate priors. The authors therefore considered alternative Markov chain Monte Carlo methods and proposed and investigated a method for posterior simulation for a variance component model based on an independence chain while using a default reference prior (Jeffrey's prior based on the restricted likelihood).

Wilson et al. (2004) more recently considered the assessment of a manufacturing process's performance when the sample of parts produced by the process is measured with error and as a result will generate measurement errors which are not of the standard $N(0, \sigma_\epsilon^2)$ form. Wilson et al. (2004)

firstly provided the standard Bayesian formulation of the one-way variance component model using a normal prior on μ , an inverse gamma (IG) prior on σ_e^2 , and the uniform shrinkage prior proposed by Daniels (1999), on σ_p^2 given σ_μ^2 . These authors secondly illustrated a model where the error variance was assumed to be proportional to the true part value. They modelled the measurements as $N(x_i, (\rho x_i)^2)$, with ρ unknown. The priors proposed on μ and σ_p^2 were normal and inverse gamma (IG) respectively, while the prior distribution assumed for ρ , was a gamma distribution. The authors then illustrated a model where the measurement errors follow a log-normal distribution. For this model the authors used the same prior distributions as for the standard Bayesian model mentioned above. In addition to these, the authors showed how to handle censored data using a truncated normal distribution, calculated tolerance intervals for the part distribution to assess the manufacturing process's performance and determined release specifications based on the producer's risk (probability of rejecting a good part) and consumer's risk (probability of accepting a bad part).

The principle purpose of the rest of this paper is to specify a Bayesian model for the data in Table 1.1 and to study the effect of departure from the usual model, which assumes that $\epsilon_{ij} \sim N(0, \sigma_e^2)$.

3. A Bayesian procedure for the Student t -distributed measurement error model

Casual inspection of the residuals shows that the underlying distribution is symmetrical, but with thick tails. To accommodate the possibility of outlying measurements, the assumption of Gaussian errors will be relaxed in the direction of the Student t -distributed family. Consider the series of independent errors $\epsilon_{ij} \mid \sigma_e^2, \lambda_{ij} \sim N(0, \frac{\sigma_e^2}{\lambda_{ij}})$ for $i = 1 \dots I$ and $j = 1 \dots J$. By placing

a prior distribution on λ_{ij} enables a wide variety of model error densities $f(\epsilon_{ij} \mid \sigma_\epsilon^2)$ to emerge as scale mixtures of normal distributions (Andrews & Mallows (1974), Carlin & Polson (1991) and Wakefield, Smith, Racine-Poon & Gelfand (1994)).

$$f(\epsilon_{ij} \mid \sigma_\epsilon^2) = \int_0^\infty p(\epsilon_{ij} \mid \sigma_\epsilon^2, \lambda_{ij}) p(\lambda_{ij}) d\lambda_{ij}.$$

For this example it is assumed that $\nu\lambda_{ij} \sim \chi_\nu^2$ so that $\epsilon_{ij} \mid \sigma_\epsilon^2 \sim t_\nu(0, \sigma_\epsilon^2)$, a Student t-distribution with mean 0, variance σ_ϵ^2 and degrees of freedom ν .

Now, the random effects model defined in equation 1 can be written as

$$\begin{aligned} y_{ij} &= \theta + \mu_i + \epsilon_{ij} \\ \text{for } i &= 1 \dots I \text{ and } j = 1 \dots J \end{aligned} \quad (2)$$

where y_{ij} is the j^{th} measured value on the i^{th} part, θ is the overall mean, μ_i represents the row or part effects where $(\mu_i \mid \sigma_p^2) \sim N(0, \sigma_p^2)$ and ϵ_{ij} is the measurement error for the j^{th} measurement on the i^{th} part where as mentioned

$$\epsilon_{ij} \mid \lambda_{ij}, \sigma_\epsilon^2 \sim N(0, \frac{\sigma_\epsilon^2}{\lambda_{ij}}) \text{ with } \nu\lambda_{ij} \sim \chi_\nu^2.$$

In matrix notation the random effects model given in equation 2 can be written as

$$\mathbf{y} = \theta \mathbf{j} + Z\boldsymbol{\mu} + \boldsymbol{\epsilon} \quad (3)$$

where

$$\mathbf{j} = [1 \ 1 \ 1 \ \dots \ 1]'$$

$$\boldsymbol{\mu} = [\mu_1 \ \mu_2 \ \mu_3 \ \dots \ \mu_I]' \text{ and } \boldsymbol{\mu} \sim N(\mathbf{0}, \sigma_p^2 I_I)$$

$$\mathbf{y} = [y_{11} \ y_{12} \ \dots \ y_{1J} \ y_{21} \ y_{22} \ \dots \ y_{IJ}]'$$

and for the example in Table 1, $I = 24$ and $J = 2$.

$$\nu_1 = I(J - 1), \nu_2 = I - 1$$

and

$$\begin{aligned}\nu_1 m_1 &= \sum_{i=1}^I \sum_{j=1}^J (y_{ij} - \bar{y}_{i.})^2 \\ \nu_2 m_2 &= J \sum_{i=1}^I (\bar{y}_{i.} - \bar{y}_{..})^2\end{aligned}$$

where

$$\begin{aligned}\bar{y}_{i.} &= \frac{1}{J} \sum_{j=1}^J y_{ij} \\ \bar{y}_{..} &= \frac{1}{IJ} \sum_{i=1}^I \sum_{j=1}^J y_{ij}\end{aligned}$$

Also

$$Z = I_I \otimes \mathbf{j}$$

where $A \otimes B$ denotes the Kronecker product of the matrices A and B . I_a is an $a \times a$ identity matrix.

An integral part of traditional Bayesian analysis is the assignment of prior distributions to the unknown parameters in the model (Van der Merwe, Pretorius, Hugo & Zellner, 2001). The choice of a prior distribution is a very difficult and controversial step in any Bayesian analysis, since the information contained in the prior distribution, which is supposed to represent what is known about the unknown parameters before the data is available, is combined with the information supplied by the data, through the likelihood function, to form the joint posterior distribution of the parameters given the data (Box & Tiao, 1973, and Gianola & Fernando, 1986).

For the prior distributions of the parameters θ , σ_p^2 , σ_e^2 , it is assumed that little is known about these parameters initially. It was therefore decided to use non-informative prior distributions to represent these parameters, i.e.

$$p(\theta, \sigma_p^2, \sigma_\epsilon^2) \propto \sigma_p^{-1} \sigma_\epsilon^{-2}$$

Since $p(\sigma_p^2) \propto \sigma_p^{-2}$ will result in improper posterior distributions, it was suggested by Gelman (2005) to use a uniform density on σ_p which is equivalent to $p(\sigma_p^2) \propto \sigma_p^{-1}$. This density can be interpreted as a limit of the half-t family on σ_p , when the scale approaches infinity (and any value of the degrees of freedom).

Another non-informative prior distribution sometimes proposed in the Bayesian literature is uniform on σ_p^2 . According to Gelman (2005) this is not recommended as it seems to have a miscalibration towards higher values and also requires $I \geq 4$ groups for a proper posterior distribution.

As prior for ν (the degrees of freedom), we assume a truncated ($\nu > 2$) exponential distribution with parameter $\xi = 0.1$. The truncation assures the finiteness of the mean and variance of the associated t-error distribution. For further details see Sahu, Dey and Branco (2003).

To obtain the joint posterior distribution, we have to multiply the likelihood function with the prior distribution. The joint posterior density is therefore given by

$$\begin{aligned} p(\theta, \sigma_\epsilon^2, \sigma_p^2, \boldsymbol{\mu}, \boldsymbol{\lambda}, \nu \mid \mathbf{y}) &\propto \left(\frac{1}{2\pi\sigma_\epsilon^2} \right)^{\frac{1}{2}(IJ+2)} \mid H \mid^{\frac{1}{2}} \\ &\exp \left\{ -\frac{1}{2\sigma_\epsilon^2} (\mathbf{y} - \theta \mathbf{j} - Z\boldsymbol{\mu})' H (\mathbf{y} - \theta \mathbf{j} - Z\boldsymbol{\mu}) \right\} \left(\frac{1}{2\pi\sigma_p^2} \right)^{\frac{1}{2}(I+1)} \exp \left\{ -\frac{1}{2\sigma_p^2} \boldsymbol{\mu}' \boldsymbol{\mu} \right\} \\ &\frac{\nu^{\frac{1}{2}\nu IJ}}{2^{\frac{1}{2}\nu IJ} \left[\Gamma \left(\frac{\nu}{2} \right) \right]^{IJ}} \left\{ \prod_{i=1}^I \prod_{j=1}^J \lambda_{ij}^{\frac{1}{2}\nu - 1} \right\} \exp \left\{ -\frac{1}{2}\nu \sum_{i=1}^I \sum_{j=1}^J \lambda_{ij} \right\} \exp(-\xi\nu) \end{aligned} \quad (4)$$

where

$$\boldsymbol{\lambda} = [\lambda_{11} \ \lambda_{12} \ \cdots \ \lambda_{1J} \ \lambda_{21} \ \lambda_{22} \ \cdots \ \lambda_{IJ}]'$$

and

$$H = \text{diag}(\lambda_{11} \ \lambda_{12} \ \cdots \ \lambda_{1J} \ \lambda_{21} \ \lambda_{22} \ \cdots \ \lambda_{IJ}).$$

4. Bayesian computation

To make appropriate inferences in a Bayesian analysis, the marginal posterior distributions and predictive densities are needed. Due to the complexity of the joint posterior distribution, however, it is impossible to obtain these marginal posterior densities analytically. It is also very difficult to obtain these marginal posterior densities numerically, due to the high number of unknowns (Van der Merwe, Pretorius & Meyer, 2003). It is therefore recommended that a Monte Carlo simulation procedure be used to estimate the marginal posterior densities of the unknown parameters and predictive densities of future observations.

Recent advances in Bayesian computation allow one to obtain a random sample from the joint posterior density of the model parameters by using a computer random number generator¹ (Wilson et al., 2004, and Wolfinger, 1998). Once a random sample is obtained, Bayesian inference can be made on any of the quality characteristics of interest. Remember, however, that this sample is dependent on the selected random number seed and that different seeds would determine different samples. If the samples are, however, large enough, differences between inferences from different samples will be small (Wolfinger, 1998).

¹ Computer-generated numbers are not really random, since computers are deterministic. But given a number to start with, generally called a random number seed, a number of mathematical operations can be performed on the seed so as to generate unrelated (pseudorandom) numbers. If the same random number seed is used more than once, identical random numbers will be generated every time. Using a different seed, would produce a different sample.

As mentioned, Markov chain Monte Carlo simulation methods can be used to obtain these random samples from the joint posterior distribution. The best known MCMC method is the Gibbs sampler which consists of repeated cycles of draws from the full conditional posterior distributions. The Gibbs sampler, however, has a principle requirement that all conditional densities must be available in the sense that random variates can be generated from them.

For the random effects model given in equation 2 and the data in Table 1, the Gibbs sampler will succeed, because the problem of dealing simultaneously with a large number of intricately related unknown parameters is reduced into a much simpler problem of dealing with one unknown quantity at a time, sampling each from its full conditional posterior distribution (Gilks, Thomas & Spiegelhalter, 1994).

One issue, however, with Gibbs sampling, referred to as the convergence of the Gibbs sampler, is whether the draws are approximately a random sample from the posterior distribution (Wilson et al. 2004). However, by running the Gibbs sampler a number of times and then discarding the draws obtained, also known as the burn-in period, the impact of initially chosen values are decreased. Also, by retaining every K^{th} draw, dependence between draws can be reduced (Wilson et al. 2004). The Gibbs sampler on the other hand also has an advantage when calculating quality characteristics, since an approximate randomly selected sample from the posterior densities of the unknown parameters is obtained after running the Gibbs sampler. This sample can then be used in a straightforward way to calculate a distribution for a particular characteristic (Wilson et al. 2004).

5. Conditional posterior distributions

As mentioned, to implement the Gibbs sampler, the full conditional posterior distributions of the unknown parameters are needed. By completing the square with respect to θ in the joint posterior distribution (equation 4) it follows that θ is normally distributed, i.e.

$$\theta \mid \mathbf{y}, \sigma_\epsilon^2, H, \boldsymbol{\mu} \sim N \left\{ \left(\frac{1}{\sigma_\epsilon^2} \mathbf{j}' H \mathbf{j} \right)^{-1} \frac{1}{\sigma_\epsilon^2} \mathbf{j}' H (\mathbf{y} - Z \boldsymbol{\mu}); \sigma_\epsilon^2 (\mathbf{j}' H \mathbf{j})^{-1} \right\}. \quad (5)$$

Also, the conditional posterior distribution of the random vector $\boldsymbol{\mu}$ is

$$\begin{aligned} \boldsymbol{\mu} \mid \theta, \sigma_\epsilon^2, \sigma_p^2, H, \mathbf{y} \sim N \left[\left(\frac{1}{\sigma_\epsilon^2} z' H z + \frac{1}{\sigma_p^2} I_I \right)^{-1} \right. \\ \left. \left(\frac{1}{\sigma_\epsilon^2} z' H \mathbf{y}^* \right); \left(\frac{1}{\sigma_\epsilon^2} z' H z + \frac{1}{\sigma_p^2} I_I \right)^{-1} \right] \quad (6) \end{aligned}$$

where $\mathbf{y}^* = \mathbf{y} - \theta \mathbf{j}$.

From equation 4 it is also clear that the conditional posterior distribution of the variance components are

$$\begin{aligned} p(\sigma_\epsilon^2 \mid \theta, \boldsymbol{\mu}, \sigma_p^2, H, \mathbf{y}) \propto (\sigma_\epsilon^2)^{-\frac{1}{2}(IJ+2)} \\ \exp \left\{ -\frac{1}{2\sigma_\epsilon^2} (\mathbf{y} - \theta \mathbf{j} - Z \boldsymbol{\mu})' H (\mathbf{y} - \theta \mathbf{j} - Z \boldsymbol{\mu}) \right\} \quad (7) \end{aligned}$$

and

$$p(\sigma_p^2 \mid \sigma_\epsilon^2, \boldsymbol{\mu}, \theta, H) \propto (\sigma_p^2)^{-\frac{1}{2}(I+1)} \exp \left\{ -\frac{1}{2\sigma_p^2} \boldsymbol{\mu}' \boldsymbol{\mu} \right\} \quad (8)$$

which are inverted Gamma distributions.

Further, the conditional posterior distribution of λ_{ij} is given by

$$p(\lambda_{ij} | \mathbf{y}, \boldsymbol{\mu}, \theta, \sigma_\epsilon^2, \sigma_p^2, \nu) \propto \lambda_{ij}^{\frac{1}{2}(\nu+1)-1} \exp \left\{ -\frac{1}{2} \lambda_{ij} \left[\nu + \frac{1}{\sigma_\epsilon^2} (y_{ij} - \theta - \mu_i)^2 \right] \right\} \quad (9)$$

which is in the general form of a Gamma distribution. Straightforward procedures can be used to simulate observations from these conditional distributions.

The conditional posterior density of ν (for $\nu > 2$) is given by

$$p(\nu | \mathbf{y}, \boldsymbol{\lambda}) \propto \frac{\nu^{\frac{1}{2}\nu IJ}}{2^{\frac{1}{2}\nu IJ} [\Gamma(\frac{\nu}{2})]^{IJ}} \prod_{i=1}^I \prod_{j=1}^J \lambda_{ij}^{\frac{1}{2}\nu} \exp \left\{ \nu \left(-\frac{1}{2} \sum_{i=1}^I \sum_{j=1}^J \lambda_{ij} - \xi \right) \right\} \quad (10)$$

Clearly, equation 10 does not correspond to any common distribution, hence we only have the kernel of the density. Geweke (2005) uses a metropolis within Gibbs step to generate random numbers from equation 10. As candidate density he uses a univariate normal distribution, with mean at the mode $\hat{\nu}$ of $p(\nu | \mathbf{y}, \boldsymbol{\lambda})$ and precision equal to $-d^2 \log p(\nu | \mathbf{y}, \boldsymbol{\lambda}) / d\nu^2 | \nu = \hat{\nu}$. This method is also used in the Bayesian Analysis Computation and Communication (BACC) software for models with Student t-distributions.

We, however, thought of using the rejection sampling method as described by Rice (1995) to simulate from $p(\nu | \mathbf{y}, \boldsymbol{\lambda})$. In order for the rejection method to be computationally efficient it was decided to use as candidate function also the normal distribution. As will be seen later, the normal distribution fits $p(\nu | \mathbf{y}, \boldsymbol{\lambda})$ so well that we could simulate directly from it.

6. Marginal posterior distributions

Recall that the data provided in Table 1 comes from a new manufacturing process and each row of Table 1 represents two measurements of iron concentration. Also recall that the two iron concentrations measured on the same part were determined by emission spectroscopy. If a part has a measurement of under 225 ppm of iron, it is considered as acceptable (Wilson et al. 2004).

Since the full conditional posterior distributions of the unknown parameters have been determined, the Gibbs sampler can be implemented to obtain the marginal posterior distributions of the random effects and variance components. MATLAB was used to do the simulations for the Gibbs sampler.

The iteration process for the **Burn-in** period was started by specifying initial values for $\sigma_\epsilon^{2(0)} = m_1 = 1721.3285$ and $\sigma_p^{2(0)} = \frac{1}{j}(m_2 - m_1) = 1693.6653$. Also, with these preselected values, we determined $\mu^{(0)}$ as follows: For part 1, the average was obtained for measurement 1 and measurement 2. The average of all the observations, $\bar{y}_{..} = \theta^{(0)}$, was then subtracted from this average. Therefore $56.63 = \left(\frac{206+258}{2}\right) - 175.37$. This was also repeated for the remaining 23 parts. Also, λ was taken as a vector of 48 chi-square random values with $\nu = 5$ degrees of freedom each.

These values were substituted into the full conditional posterior densities of the unknown parameters to draw $\nu^{(1)}, \theta^{(1)}, \sigma_\epsilon^{2(1)}, \sigma_p^{2(1)}, \mu^{(1)}$ and $\lambda^{(1)}$. The full conditional posterior densities were updated after every iteration. The results from the burn-in period were obtained after running the procedure 30 000 times. Every 10^{th} value for $\nu, \theta, \sigma_\epsilon^2, \sigma_p^2, \mu$ and λ was stored, meaning that 3 000 values were stored for each of the unknown parameters. Of these 3 000 stored combinations, the first 2 999 stored combinations were disregarded and the values stored on the $30\,000^{th}$ iteration, i.e. $\nu^{(3000)}, \theta^{(3000)}, \sigma_\epsilon^{2(3000)},$

$\sigma_p^{2(3000)}$, $\mu^{(3000)}$ and $\lambda^{(3000)}$, were used as starting values for the Gibbs sampling procedure.

The new starting values obtained from the burn-in period were then substituted into the full conditional posterior densities and the Gibbs sampling procedure was run 100 000 times. Again, every 10th sample was saved, meaning that the total number of samples saved was 10 000. Also, the full conditional posterior densities were updated after every iteration. These results obtained from the Gibbs sampler were then used to obtain the histograms of the marginal posterior distributions $p(\nu | \mathbf{y})$, $p(\theta | \mathbf{y})$, $p(\sigma_\epsilon^2 | \mathbf{y})$, $p(\sigma_p^2 | \mathbf{y})$ and for example $p(\lambda_{20} | \mathbf{y})$.² Estimates from the marginal posterior densities of functions of the variance components such as the total variance $\sigma_\epsilon^2 + \sigma_p^2$ and variance ratio $\sigma_\epsilon^2 / (\sigma_\epsilon^2 + \sigma_p^2)$ could also easily be obtained. The posterior distributions are displayed in Figures 2–7.

Different starting values for the variance components were also considered for the iteration process for the burn-in period. For example, for $\sigma_\epsilon^{2(0)}$, 500 and 2500 were used, while 400 and 2400 were used for $\sigma_p^{2(0)}$. The marginal posterior distributions obtained from the Gibbs sampler were, however, for all practical purposes the same as those illustrated.

From Figure 1 it can be seen that the conditional posterior density of ν appears to be symmetrical and bell-shaped. As mentioned, it was decided to approximate it with a normal distribution.

From Figure 1 it is clear that the posterior mode is equal to 3, indicating that the most frequently used number of degrees of freedom to simulate the variance components and random effects, was 3. The low number of degrees of freedom is an indication that a t -distribution will fit the measurement errors better than the normal distribution.

² The marginal posterior densities for $p(\lambda_i | y) \ \forall \ i = 1 \cdots 24$ can also easily be obtained. $i = 20$ was selected for illustrative purposes only.

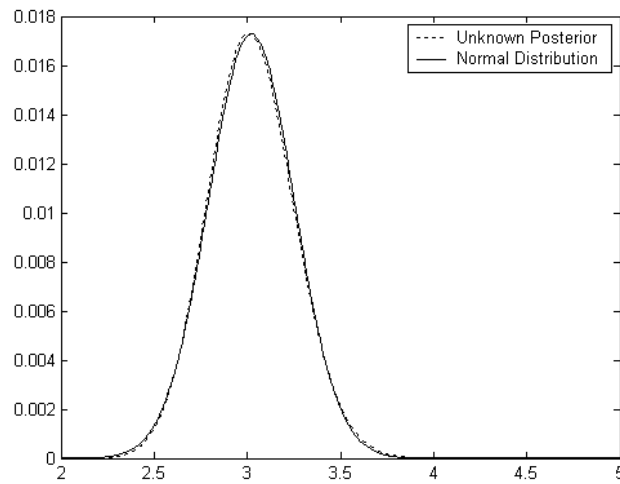


Figure 1 Unknown posterior density of the degrees of freedom ν and a normal distribution

As expected, the histogram of the marginal posterior distribution of θ , Figure 2, is symmetrical due to the fact that the conditional posterior distribution of θ , equation (5), is normal. Figure 2 compares well with the posterior distributions of the production mean which are illustrated in Figure 2 of Wilson et al. (2004).

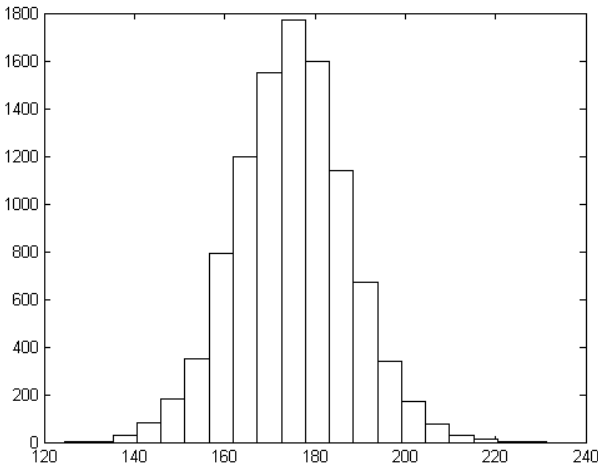


Figure 2 Histogram of the marginal posterior density of θ

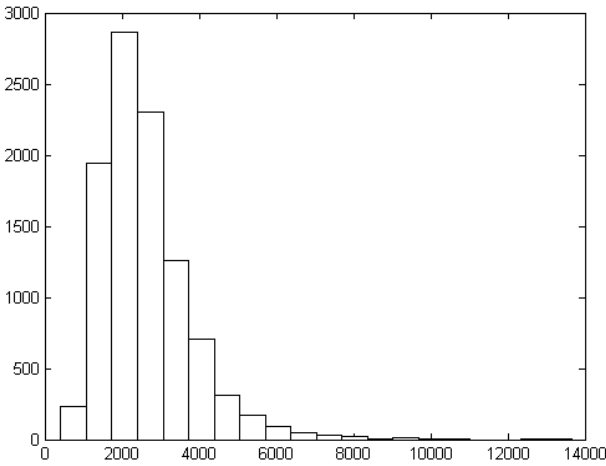


Figure 3 Histogram of the marginal posterior density of σ_ϵ^2

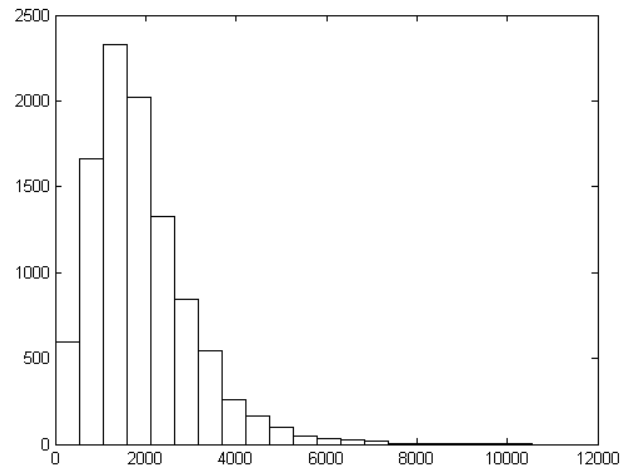


Figure 4 Histogram of the marginal posterior density of σ_p^2

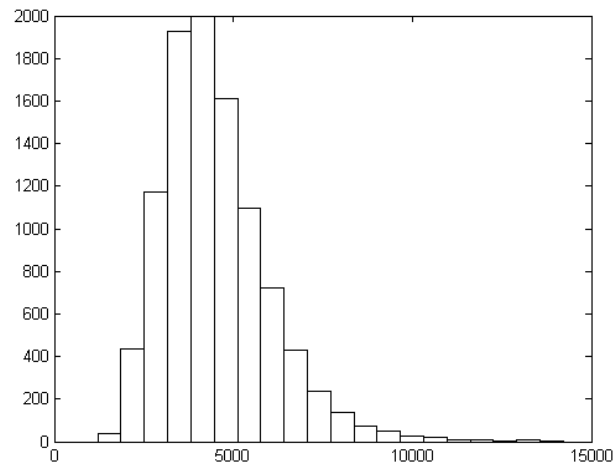


Figure 5 Histogram of the marginal posterior density of $(\sigma_\epsilon^2 + \sigma_p^2)$

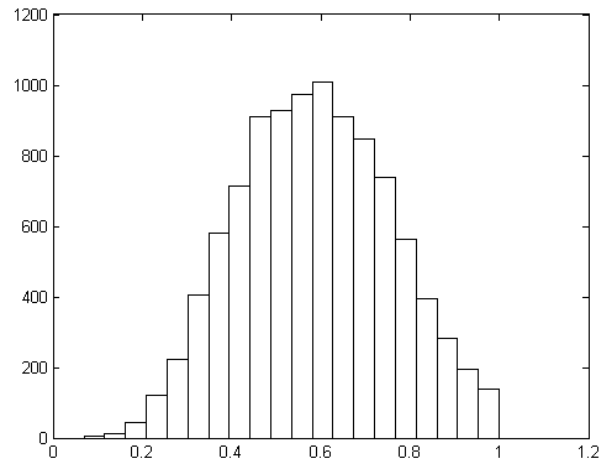


Figure 6 Histogram of the marginal posterior density of $\sigma_\epsilon^2 / (\sigma_\epsilon^2 + \sigma_p^2)$

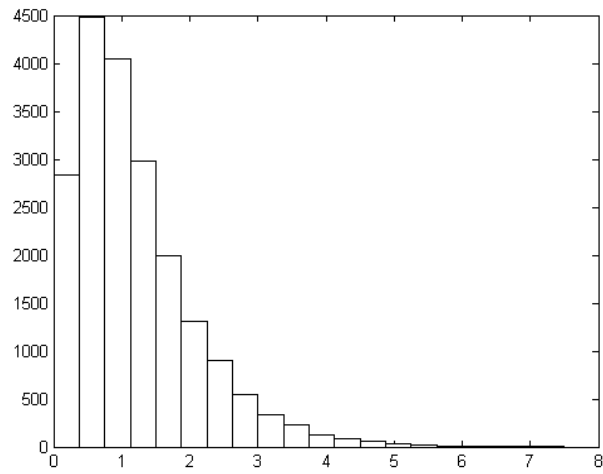


Figure 7 Histogram of the marginal posterior density of λ for the 20th day

As expected, the histogram of the marginal posterior distribution of θ , Figure 2, is symmetrical due to the fact that the conditional posterior distribution of θ , equation (5), is normal. Figure 2 compares well with the posterior distributions of the production mean which are illustrated in Figure 2 of Wilson et al. (2004).

It should be noted that the histogram of the marginal posterior distribution of σ_ϵ^2 (measurement variance) is usually symmetrical or fairly symmetrical due to the high number of degrees of freedom usually associated with it. If one, however, considers Figure 3 it can be seen that the histogram of σ_ϵ^2 is relatively negatively skewed. The reason for this is the low number of degrees of freedom associated with σ_ϵ^2 (i.e. $I(J - 1) = 24$) due to the small size of 24 selected parts with only 2 measurements per part.

The histogram of the marginal posterior distribution of σ_p^2 is similar to the graphs of the posterior distributions of the production variance displayed by Wilson et al. (2004) (Figure 2) for the normal, multiplicative and log-normal measurement error cases.

For comparative purposes the above simulations were also repeated for cases where the degrees of freedom were held fixed at 3, 5, 10, 15, 20, 25 and 30 and for the case where the residuals were considered to be random $N(0, \sigma_\epsilon^2)$ variables. In Table 2 the estimated posterior median values for the variance components, functions of the variance components and λ_{20} are given.

Comparing the results given in Table 2, reveal that the overall mean θ , remains more or less constant for all cases mentioned. The same applies to the variance component measuring the between parts variation, σ_p^2 . This is to be expected since neither θ nor σ_p^2 depends directly on the degrees of freedom ν .

Table 2 Estimated posterior median values for variance components, functions of variance components and λ_{20}

Posterior median for	ν Simulated	$\nu=3$	$\nu=5$	$\nu=10$	
θ	174.93	175.14	175.01	175.15	
σ_ϵ^2	2394.90	2393.50	2168.90	2024.50	
σ_p^2	1678.40	1676.40	1666.90	1699.30	
$\sigma_\epsilon^2 + \sigma_p^2$	4272.40	4310.10	4011.30	3863.40	
$\sigma_\epsilon^2 / (\sigma_\epsilon^2 + \sigma_p^2)$	0.5849	0.5855	0.5598	0.5390	
λ_{20}	0.9930	1.0052	0.9865	0.9936	

Posterior median for	$\nu=15$	$\nu=20$	$\nu=25$	$\nu=30$	$\epsilon_{ij} \sim N(0, \sigma_\epsilon^2)$
θ	175.28	175.37	175.42	175.196	175.193
σ_ϵ^2	1935.90	1911.70	1893.50	1902.20	1823.30
σ_p^2	1720.50	1716.10	1720.20	1719.80	1730.70
$\sigma_\epsilon^2 + \sigma_p^2$	3802.50	3739.50	3728.00	3720.90	3659.70
$\sigma_\epsilon^2 / (\sigma_\epsilon^2 + \sigma_p^2)$	0.5228	0.5218	0.5171	0.5172	0.5071
λ_{20}	0.9953	0.9954	0.9976	0.9996	—

Also, even though the simulation of λ_{ij} depends on the degrees of freedom, the posterior median values for λ_{20} , remained approximately the same for all the cases mentioned in Table 2. Also, recall that when ν was simulated, the estimated posterior mode of the marginal density of ν was equal to 3, indicating that the most frequently used number of degrees of freedom was 3. From Table 2 it is therefore evident that as the degrees of freedom increase and the student t -distribution thus approaches a normal distribution, the estimated posterior median values for σ_ϵ^2 decrease and approach the estimated posterior median value of σ_ϵ^2 for the case where neither ν nor λ is present and where it is assumed that the residuals follow a $N(0, \sigma_\epsilon^2)$ distribution. The same can be seen for the estimated posterior median values for the total variance, $\sigma_\epsilon^2 + \sigma_p^2$,

Table 3 95% Equal tails credibility intervals and interval widths of quantities

Quantity of interest	ν -Simulated		$\nu = 3$	
	95% CI	Width	95% CI	Width
θ	149.8731 — 200.0634	50.1903	150.8202 — 200.0941	49.2738
σ_ϵ^2	1092.7 — 5554.6	4461.8	1080.3 — 5587.4	4507.1
σ_p^2	301.1450 — 4712.3	4411.2	298.4185 — 4662.8	4364.4
σ_ϵ^2	2284.2 — 8086.0	5801.8	2334.2 — 8141.0	5806.8
$\sigma_\epsilon^2 / (\sigma_\epsilon^2 + \sigma_p^2)$	0.2742 — 0.9253	0.6511	0.2650 — 0.9268	0.6618
λ_{20}	0.1397 — 3.4833	3.3436	0.1453 — 3.4058	3.2604

Quantity of interest	$\nu = 30$		$\epsilon_{ij} \sim N(0, \sigma_\epsilon^2)$	
	95% CI	Width	95% CI	Width
θ	153.4712 — 197.1296	43.6583	153.1817 — 197.0485	43.8668
σ_ϵ^2	1060.5 — 3716.5	2656.0	1067.1 — 3472.1	2405.1
σ_p^2	246.6363 — 4342.6	4095.9	301.9871 — 4320.9	4018.9
σ_ϵ^2	2366.7 — 6412.7	4046.0	2397.0 — 6276.1	3879.1
$\sigma_\epsilon^2 / (\sigma_\epsilon^2 + \sigma_p^2)$	0.2536 — 0.9253	0.6716	0.2494 — 0.9103	0.6610
λ_{20}	0.5752 — 1.5866	1.0114	— — —	—

and the variance ratio, $\sigma_\epsilon^2/(\sigma_\epsilon^2 + \sigma_p^2)$. This is also to be expected since both of these quantities are functions of the error variance component, σ_ϵ^2 .

By considering Table 3, it can be seen that the estimated interval width for the location parameter θ decreases generally as the degrees of freedom increase and the Student t -distribution thus approaches a normal distribution. This is to be expected since the Student t -distribution is wider than a Normal distribution. The same can be seen for σ_ϵ^2 , $\sigma_\epsilon^2 + \sigma_p^2$, $\sigma_\epsilon^2/(\sigma_\epsilon^2 + \sigma_p^2)$ and λ_{20} . Although the interval width also decreases slightly for σ_p^2 , this decrease is not as dramatic as for σ_ϵ^2 .

As in Wilson et al. (2004), we will also look at tolerance intervals in the case of non-standard measurement errors (t -distributed). This will be illustrated in the next section.

7. Tolerance intervals

In any production process, designers will specify tolerances for various characteristics based on considerations of requirements for fit or function in use or in subsequent levels of assembly. The dimensions within which a produced part should fall in order to be acceptable, is a typical example (Easterling, Johnson, Bement & Nachtsheim, 1991). To protect against measurement error and to keep the production facility on its toes, designers sometimes specify tolerance limits with an interval width less than the width of the true required tolerance limits. Since these ad hoc tolerances may impose undue costs due to scrap or rework, it is desirable to take a more systematic look at the determination of tolerances, taking measurement error into account (Easterling et al. 1991).

Based on this information and since the Bayesian model has been specified and the variance components and random effects have been determined, some

important research questions can now be asked about the data given in Table 1. These are as follows:

1. Assuming the manufacturing process is in control, can a limit t be found such that 90% of future measurements are greater than t with 95% probability?
2. Can an interval (t_l, t_u) be constructed so that a new observation from one of the original parts will fall in (t_l, t_u) with 95% probability?
3. What fraction of the process measurements lie above some preselected specification limit s , and how much uncertainty is associated with this estimated fraction?

These three questions, and many similar ones, are addressed by Bayesian tolerance intervals which allow inference about the quantiles of a probability distribution that adequately describes a process (Wolfinger, 1998).

Three kinds of commonly used tolerance intervals address respectively these three research questions. These are:

1. The (α, δ) *tolerance interval*, where α is the content (the proportion of the population to be contained by the interval) and δ is the confidence (the reliability of the interval). Both α and δ lie between 0 and 1 and are typically assigned values of 0.90, 0.95 or 0.99.
2. The α -*expectation tolerance interval*, where α is the expected coverage of the interval. Again, α is measured on a probability scale and is typically set to a value close to 1. In contrast to the (α, δ) tolerance intervals, the α -expectation intervals focus on prediction of one or a few future observations from the process and consequently tend to be narrower than the corresponding (α, δ) intervals.
3. The *fixed-in-advance tolerance interval*, in which the interval is constant and one wishes to estimate the proportion of process measurements it contains. Fixed-in-advance intervals invert the prediction problem by considering the content of predetermined bounds.

All three kinds of tolerance intervals can take the following forms: lower limit (t_l, ∞) , an upper limit (∞, t_u) , or a two-sided limit (t_l, t_u) . For further details about confidence intervals and tolerance limits see Hahn and Meeker (1991) and Wolfinger (1998).

As mentioned, these tolerance intervals will address the statistical problem of inference about the quantiles of a probability distribution that is assumed to adequately describe a process (Van der Merwe & Hugo, 2007).

The construction of tolerance intervals has a rich history dating back over half a century (Wolfinger, 1998). For reviews on this research see Wilks (1941), Wald (1942), Guttman (1970), Zacks (1971), Mee and Owen (1983), Mee (1984a), Miller (1989), Hahn and Meeker (1991), Bhaumik and Kulkarni (1991, 1996) and Vangel (1992). More recently, Wolfinger (1998) proposed the estimation of tolerance intervals using a one-way variance component model and Bayesian simulation. Wolfinger (1998) also pointed out that the frequentist analysis of tolerance intervals can become quite complex, even for balanced one-way random effects models. Furthermore, the frequentist analysis differs depending on the kind of tolerance interval and particular model under consideration. Also, based on work done by Weerahandi (1993) (see also Weerahandi, 1995), Krishnamoorthy and Mathew (2004) introduced one-sided tolerance limits for balanced and unbalanced one-way random models using the generalized confidence interval approach.

A simulation-based approach will now be presented for determining Bayesian tolerance intervals for the one-way variance component model given in equation 3 with non-standard measurement errors. The procedure will then be applied to the data given in Table 1.

7.1 α -expectation tolerance interval

According to Wolfinger (1998), the α -expectation tolerance intervals address research question 2 and focus on prediction of one or a few future observations from the process.

By using the results given in equations 5–10 the following theorems can now be proved.

Theorem 7.1.1

The predictive density of the average of q future measurements $y_{f1}, y_{f2}, \dots, y_{fq}$ for a new or unknown part, given the variance components, $\lambda_{fj^*} (j^* = 1, \dots, q)$ and θ is normally distributed with mean

$$E(\bar{y}_f \mid \theta, \sigma_\epsilon^2, \sigma_p^2, \lambda_{f1}, \dots, \lambda_{fq}) = \theta \quad (11)$$

and variance

$$Var(\bar{y}_f \mid \theta, \sigma_\epsilon^2, \sigma_p^2, \lambda_{f1}, \dots, \lambda_{fq}) = \frac{\sigma_\epsilon^2}{q^2} \sum_{j^*=1}^q \frac{1}{\lambda_{fj^*}} + \sigma_p^2 \quad (12)$$

Proof

The proof is given in the appendix.

Theorem 7.1.2

The predictive density for the average of q future measurements $\tilde{y}_{i1}, \tilde{y}_{i2}, \dots, \tilde{y}_{iq}$ for a specific part (i^{th} part) or a new part similar to the given specific part, given the variance components, $\lambda_{i11}, \lambda_{i12}, \dots, \lambda_{i1q}, \lambda_{i21}, \lambda_{i22}, \dots, \lambda_{i2q}$ and θ is normally distributed with mean

$$E(\bar{\tilde{y}}_i \mid \theta, u_i, \lambda_{i11}, \dots, \lambda_{i2q}, \sigma_\epsilon^2, \sigma_p^2) = \theta + u_i \quad (13)$$

and variance

$$\begin{aligned} & Var(\bar{y}_{i.} \mid \theta, u_i, \lambda_{i11}, \dots, \lambda_{i2q}, \sigma_\epsilon^2, \sigma_p^2) \\ &= \frac{\sigma_\epsilon^2}{q^2} \left\{ p \sum_{j^*=1}^q \frac{1}{\lambda_{i1j^*}} + (1-p) \sum_{j^*=1}^q \frac{1}{\lambda_{i2j^*}} \right\} \end{aligned} \quad (14)$$

Proof

The proof is given in the appendix.

To be able to provide comparative results with results obtained by Wilson et al. (2004), all future posterior densities and tolerance intervals will be based on a $N(\theta, \sigma_p^2)$ distribution which represents the variation amongst the parts. Figure 8 represents the unconditional predictive density of x , the parts distribution which conditional on $\theta, \sigma_\epsilon^2, \sigma_p^2, \boldsymbol{\mu}, \boldsymbol{\lambda}$, is distributed $N(\theta, \sigma_p^2)$.

The Bayesian simulation for obtaining Figure 8 was performed in the following way:

1. By using the Gibbs sampling procedure, the variance components and random effects were generated from the joint posterior distribution in the case where the degrees of freedom were simulated. This was also done for the cases where the degrees of freedom were held fixed.
2. For each set of simulated values $\theta, \sigma_\epsilon^2, \sigma_p^2, \boldsymbol{\mu}, \boldsymbol{\lambda}$ for the case where ν was simulated, the conditional predictive part distribution is normal with mean θ and variance σ_p^2 .
3. For the data in Table 1, steps 1 and 2 were repeated $k = 10\,000$ times. Using the Rao-Blackwell argument (see Gelfand and Smith, 1991), density estimates of the unconditional predictive part distribution were obtained by averaging the conditional predictive densities over the $k = 10\,000$ repetitions.

For comparative purposes, the median values and 95% credibility intervals for the predictive part distribution, for $\nu = 3, \nu = 5, \nu = 10, \nu = 15, \nu = 20$,

$\nu = 25$, $\nu = 30$, ν simulated and the case where $\epsilon_{ij} \sim N(0, \sigma_\epsilon^2)$ are given in Table 4.

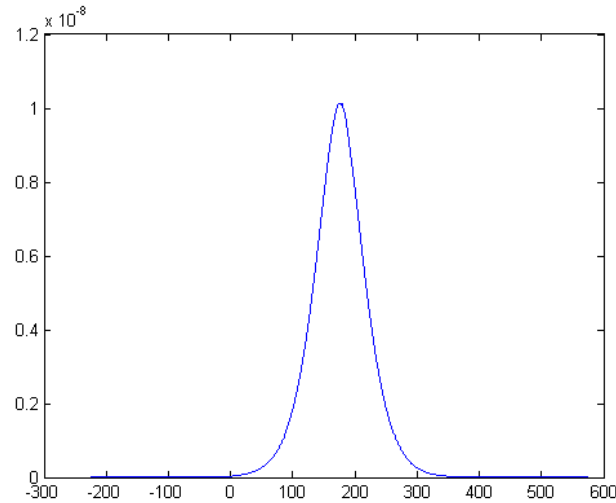


Figure 8 Unconditional predictive part distribution, x , if ν is simulated

Table 4 Median values and 95% equal tails credibility intervals for the predictive part distribution

Degrees of Freedom	Median	95% Upper Prediction Limit
ν —Simulated	176.0217	249.1971
$\nu = 3$	175.5878	251.9463
$\nu = 5$	174.4706	247.1394
$\nu = 10$	175.3312	250.3784
$\nu = 15$	175.5526	249.9964
$\nu = 20$	175.1292	247.8211
$\nu = 25$	175.3357	249.2385
$\nu = 30$	175.3299	246.9510
$\epsilon_{ij} \sim N(0, \sigma_\epsilon^2)$	174.9438	248.7445

It can be seen from Table 4, that, as expected, the median values for these distributions are for all practical purposes equal. Also note that the median values are for all practical purposes equal to 175.37, the average of the data

given in Table 1. It can also be seen that the predictive part distribution compares well with the predicted parts plot in Figure 2 on page 199 of Wilson et al. (2004).

For illustrative purposes, using results obtained if the number of degrees of freedom ν is simulated, it follows that the process is in control if 95% or more of the part measurements fall below 249.1971.

7.2 One-sided (α, δ) tolerance intervals

Assuming that a process is in a state of control, research question 1 can be addressed by the (α, δ) tolerance intervals. These (α, δ) tolerance intervals are typically applied in cases requiring long-run prediction about numerous observations from this in-control process and inference is based on the actual quantiles of the assumed underlying probability distribution (Wolfinger, 1998). Based on available measurements, manufacturers often use these (α, δ) tolerance intervals to predict the future performance of a product (Wolfinger, 1998).

To construct one-sided (α, δ) tolerance intervals, the estimated marginal posterior density of q^* must be obtained. Consider for example the $(1 - \alpha)^{th}$ quantile of the $N \left\{ (\theta + \mu_i); \frac{\sigma_\epsilon^2}{q^2} \left[p \sum_{j^*=1}^q \frac{1}{\lambda_{i1j^*}} + (1 - p) \sum_{j^*=1}^q \frac{1}{\lambda_{i2j^*}} \right] \right\}$ distribution.

The average of future data of the i^{th} part or a part similar to the i^{th} part is described by this distribution. The Bayesian simulation procedure for obtaining the posterior distribution of q^* can be performed as follows:

1. Use the Gibbs sampling procedure (5 – 10) to generate the variance components, θ , λ and random effects from the joint posterior distribution for the cases where ν was held fixed and for the case where ν was simulated.

2. For given values of $\theta, \sigma_\epsilon^2, \sigma_p^2, \mu$ and λ , determine³

$$q^* = (\theta + \mu_i) - z_\alpha \left\{ \frac{\sigma_\epsilon^2}{q^2} \left[p \sum_{j^*=1}^q \frac{1}{\lambda_{i1j^*}} + (1-p) \sum_{j^*=1}^q \frac{1}{\lambda_{i2j^*}} \right] \right\}^{\frac{1}{2}}.$$

3. Repeat steps 1 and 2 $k = 10\,000$ times to obtain the estimated marginal posterior density of q^* .

For comparative purposes, the histogram of the empirical posterior distribution of the 0.95th quantile of the parts distribution is displayed in Figure 9. The histogram in Figure 9. is obtained by calculating

$$q_p^* = \theta + 1.645 \sigma_p$$

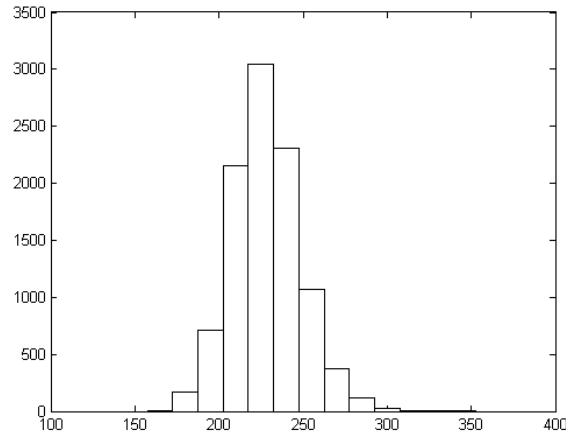
using the Gibbs sampled values for θ and σ_p^2 . The (0.95, 0.90) one sided upper tolerance bound can then easily be obtained from the posterior distribution of the 0.95th quantile by ranking the simulated q_p^* values and obtaining the 0.90th percentile of these ranked simulated values. The one sided upper tolerance bounds for $\nu = 3, \nu = 5, \nu = 10, \nu = 15, \nu = 20, \nu = 25, \nu = 30, \nu$ simulated and the case where $\epsilon_{ij} \sim N(0, \sigma_\epsilon^2)$, are given for comparative purposes in Table 5.

The credibility limit for the Normal distribution is only somewhat smaller than the corresponding ones for the t -distribution. Also, the histogram in Figure 9 compares very well with the posterior distributions of the 0.95th quantile of the part distribution (especially the Normal and log-Normal measurement error models) given in Figure 3 on page 200 of Wilson et al. (2004)

³ Remember since $J = 2$ measurements were taken on the same part, q^* needs to be determined twice, once for each λ_{ij} for $j = 1, 2$.

Table 5 90% upper credibility limits of the 0.95th quantile of the part distribution

Case	Upper Credibility Limits
ν —Simulated	263.5571
$\nu = 3$	263.4439
$\nu = 5$	262.4140
$\nu = 10$	261.9401
$\nu = 15$	262.0021
$\nu = 20$	261.1025
$\nu = 25$	261.1841
$\nu = 30$	261.1790
$\epsilon_{ij} \sim N(0, \sigma_\epsilon^2)$	261.0426

**Figure 9** Histogram of the posterior distribution of the 0.95th quantile of the parts distribution

The histograms of the marginal posterior densities of

1. $q_l^* = (\theta + \mu_i) - 1.96 \left\{ \frac{\sigma_\epsilon^2}{q^2} \left[p \sum_{j^*=1}^q \frac{1}{\lambda_{i1j^*}} + (1-p) \sum_{j^*=1}^q \frac{1}{\lambda_{i2j^*}} \right] \right\}^{\frac{1}{2}}$
and
2. $q_u^* = (\theta + \mu_i) + 1.96 \left\{ \frac{\sigma_\epsilon^2}{q^2} \left[p \sum_{j^*=1}^q \frac{1}{\lambda_{i1j^*}} + (1-p) \sum_{j^*=1}^q \frac{1}{\lambda_{i2j^*}} \right] \right\}^{\frac{1}{2}},$
the $[(1 \pm 0.95)/2]^{th}$ quantiles of the marginal posterior density of q^* for

the average of q measurements on the i^{th} part or a similar future part, as well as

3. $q_l^* = \theta - 1.96 \left\{ \frac{\sigma_p^2}{q^2} \sum \frac{1}{\lambda_{fj^*}} + \sigma_p^2 \right\}^{\frac{1}{2}}$ and
4. $q_u^* = \theta + 1.96 \left\{ \frac{\sigma_p^2}{q^2} \sum \frac{1}{\lambda_{fj^*}} + \sigma_p^2 \right\}^{\frac{1}{2}}$, the $(1 \pm 0.95)^{th}$ quantiles of the predictive density of q^* for the average of q future measurements of a new or unknown part, can also easily be determined by using steps 1 to 3 described above. Remember that for 1 and 2 above, q_l^* and q_u^* need to be determined twice, once for each λ_{ij} for $j = 1, 2$. Also, for 3 and 4 above, $\lambda_{fj^*} (j^* = 1, \dots, q)$ are simulated from the prior distribution of λ . For illustrative purposes, Figure 10, displays the histograms of 5 and 6 where
5. $q_l^* = \theta - 1.96 \left\{ \sigma_p^2 \right\}^{\frac{1}{2}}$ and
6. $q_u^* = \theta + 1.96 \left\{ \sigma_p^2 \right\}^{\frac{1}{2}}$, the posterior distribution of the quantiles for the part distribution.

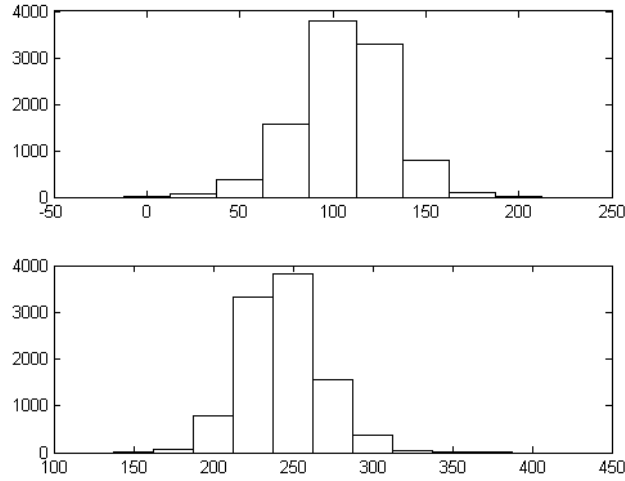


Figure 10 Histogram of the $[(1 \pm 0.95)/2]^{th}$ quantiles of the posterior density for the part distribution

7.3 Two-sided (α, δ) tolerance intervals

In some cases a two-sided interval is needed that is one-dimensional and symmetric about the posterior mean or some other form of central tendency, Wolfinger (1998) mentioned that constructing these two-sided (α, δ) tolerance intervals are slightly more complex. The simple procedure of computing upper and lower limits separately and then combining them, is not precisely valid, since these two quantiles, i.e. q_l^* and q_u^* , do not have a posterior correlation of 1.

Wolfinger (1998) therefore suggested that one way of constructing a valid two-sided (α, δ) tolerance interval, is to begin by computing the two quantities for each of the sampled values

$$q_l^* = \theta - z_{[(1+\alpha)/2]} \{\sigma_p^2\}^{\frac{1}{2}} \text{ and}$$

$$q_u^* = \theta + z_{[(1+\alpha)/2]} \{\sigma_p^2\}^{\frac{1}{2}}$$

for the marginal posterior density of the true part distribution. These (q_l^*, q_u^*) pairs then form a sample from the bivariate posterior density of the $[(1 + \alpha)/2]^{th}$ and $[(1 - \alpha)/2]^{th}$ quantiles. Bayesian confidence regions for these bivariate samples can be obtained, but are difficult to use in practice, since they are two-dimensional ellipsoids. Wolfinger (1998), however, succeeded in constructing a two-sided (α, δ) tolerance interval that is one-dimensional and symmetric about the mean. To obtain such an interval, Wolfinger (1998) suggested to first form a scatter plot of q_l^* versus q_u^* , with q_l^* on the vertical axis. One must then proceed by constructing the reference line given by

$$q_l^* = -q_u^* + 2\bar{y}_{..}$$

Then draw two additional lines, one parallel to each axis and intersecting on the reference line. This intersecting point is then slid along the reference

line until $100(1 - \delta)\%$ of the (q_l^*, q_u^*) pairs are contained in the half-rectangle opening towards the lower right portion of the graph. The coordinates of the resulting intersection point form a two-sided (α, δ) tolerance interval of the desired form. This procedure is graphically illustrated for the data in Table 1.1, for the 0.95^{th} quantile of the part distribution with ν simulated, in Figure 11.

In Table 6, the two-sided $(0.90, 0.95)$ tolerance intervals for the 0.95^{th} quantile of the part distribution are given for $\nu = 3, \nu = 5, \nu = 10, \nu = 15, \nu = 20, \nu = 25, \nu = 30, \nu$ simulated and the case where $\epsilon_{ij} \sim N(0, \sigma_\epsilon^2)$.

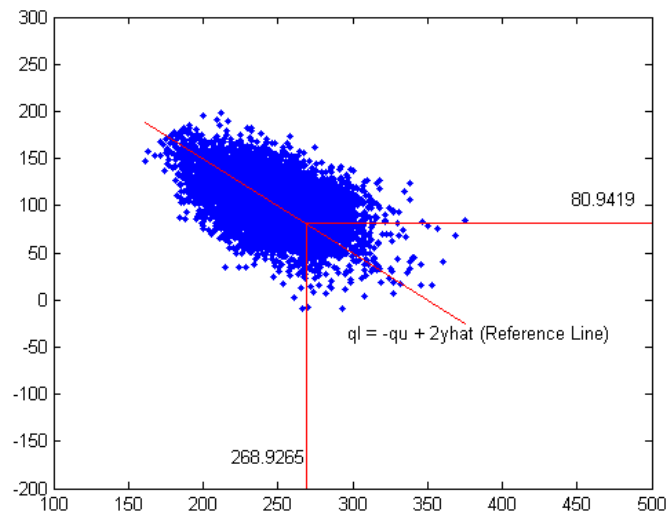


Figure 11 Illustration of a two-sided $(0.90, 0.95)$ tolerance interval for the part distribution

Table 6 Two-sided (0.90, 0.95) tolerance intervals for different cases for the data in Table 1

Case	Part Distribution	Interval Width
ν —Simulated	80.9419 – 268.9265	187.9847
$\nu = 3$	81.2023 – 269.0781	187.8758
$\nu = 5$	81.4403 – 268.5799	187.1396
$\nu = 10$	82.9635 – 267.3421	184.3786
$\nu = 15$	81.4668 – 269.0902	187.6234
$\nu = 20$	82.9109 – 267.8228	184.9119
$\nu = 25$	82.5747 – 268.2573	185.6826
$\nu = 30$	82.9389 – 267.4530	184.5141
$\epsilon_{ij} \sim N(0, \sigma_\epsilon^2)$	83.6694 – 266.7155	183.0461

For illustrative purposes, the two-sided (0.90, 0.95) tolerance interval for the part distribution with ν simulated, given by (80.9419 – 268.9265), can be interpreted as follows: 90% of emission spectroscopy true values of iron in parts per million will fall in the interval (80.9419 – 268.9265) with 95% probability.

According to Wilson et al. (2004), the 95% upper tolerance bound to contain 95% of the population is 286.7 for the normal model, 277.6 for the log-normal model and 305.3 for the censored normal model. These authors also mentioned that the upper tolerance bound for the censored normal model will be higher, since the distributions for this model are somewhat wider due to the additional uncertainty introduced when exact measurements are not made. If the upper limits of the two-sided (0.90, 0.95) tolerance intervals for the true part measurement are compared to these results obtained by Wilson et al. (2004), it can be seen that for the case assuming that ν is simulated, 268.9265, compares very well with the results obtained by Wilson et al. (2004). This is also true for other degrees of freedoms used (see table 6).

7.4 Fixed-in-advance tolerance intervals

Fixed-in-advance tolerance intervals invert the prediction problem by considering the content of predetermined bounds, and according to Wolfinger (1998), provides an answer to research question 3.

To determine the content of a fixed-in-advance tolerance interval, the posterior density of this content has to be determined. For example, suppose an upper fixed-in-advance limit of s is specified for data assumed to arise from a new batch. Then, for the average of q future measurements on the i^{th} part or future part similar to the i^{th} part, compute

$$c^* = 1 - \Phi \left[\frac{s - (\theta + \mu_i)}{\sqrt{\frac{\sigma_\epsilon^2}{q^2} \left[p \sum_{j^*=1}^q \frac{1}{\lambda_{i1j^*}} + (1-p) \sum_{j^*=1}^q \frac{1}{\lambda_{i2j^*}} \right]}} \right]$$

where c^* has to be determined twice, once for each λ_{ij} for $j = 1, 2$. Also, for the true part measurements, compute

$$c^* = 1 - \Phi \left[\frac{s - \theta}{\sqrt{\sigma_p^2}} \right].$$

It must be noted that for both of the above cases, $\Phi[\cdot]$ represents the standard normal cumulative distribution function. Therefore, a content c^* is found for the true part measurements.

To illustrate, suppose that a lower fixed-in-advance limit of $s = 225$ is specified for the data given in Table 1. This limit is selected, since according to Wilson et al. (2004), a part is considered to be acceptable if it has under 225 ppm of iron. By selecting $s = 225$ as lower fixed-in-advance limit, the content of the interval $(225, \infty)$ is determined for each observation in the sample.

In other words, we are estimating the posterior density of the content, or the proportion of parts produced by the process, that is not acceptable.

In Figure 12 the histogram of the posterior density of the content of the true part measurements is displayed for the manufacturing process with a fixed-in-advance lower limit equals to 225 ppm of iron and ν simulated.

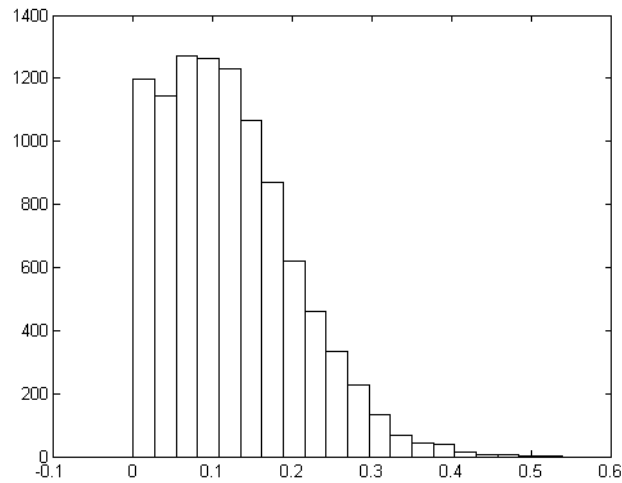


Figure 12 Histogram of posterior density of the content of a fixed in advance lower limit $s = 225$ ppm of iron for the true part measurements with ν simulated

From Figure 12 it can be seen that the posterior contents are negatively skewed and has a posterior median equals to 0.1098. This means that if the number of degrees of freedom ν is simulated, on average 10.98 % unacceptable parts will be produced by the manufacturing process with 95% probability.

For comparative purposes, the posterior median values and 95% equal tails credibility intervals for the content of the fixed-in-advance lower limit equal to $s = 225$ ppm of iron, for the true part measurements, are given in Table 7 for

the cases where $\nu = 3, \nu = 5, \nu = 10, \nu = 15, \nu = 20, \nu = 25, \nu = 30, \nu$ simulated and the case where $\epsilon_{ij} \sim N(0, \sigma_\epsilon^2)$.

From Table 7 it can be seen that the 95% credibility intervals of the content of the fixed-in-advance lower limit $s = 225$ ppm of iron for the true part measurements tend to be narrower and the posterior median values tend to be higher as the degrees of freedom increase.

Table 7 Posterior median values and 95% equal tails credibility intervals for the fixed-in-advance lower limit $s = 225$ ppm of iron for different cases for the data in Table 1

Case	Median	True Part Measurements	Interval Width
ν -Simulated	0.1098	0.0013 – 0.3083	0.3070
$\nu = 3$	0.1094	0.0011 – 0.3148	0.3137
$\nu = 5$	0.1083	0.0014 – 0.3049	0.3035
$\nu = 10$	0.1100	0.0014 – 0.3020	0.3006
$\nu = 15$	0.1128	0.0012 – 0.2967	0.2954
$\nu = 20$	0.1120	0.000969 – 0.2976	0.2967
$\nu = 25$	0.1132	0.0005385 – 0.2945	0.2939
$\nu = 30$	0.1122	0.0005163 – 0.2942	0.2937
$\epsilon_{ij} \sim N(0, \sigma_\epsilon^2)$	0.1123	0.0018 – 0.2977	0.2960

8. Checking for outliers

In the previous sections, the assumption of Gaussian errors was relaxed in the direction of the Student t -family to accommodate for the possibility of outlying part measurements. According to Wakefield et al. (1994) the scale parameter λ_{ij} or $\log_{10}(\lambda_{ij})$ serves as a good indicator to detect outlying part measurements, since there is a λ_{ij} for each measurement on each part. The prior expectation of $\log_{10}(\lambda_{ij})$ is 0, so that a $\log_{10}(\lambda_{ij})$ -value substantially below zero indicates that the j^{th} measurement on the i^{th} part is likely to be an

outlier (Du Plessis & Van der Merwe, 1996). Interval estimates of $\log_{10}(\lambda_{ij})$ can also easily be constructed.

In Table 8 the 90% credibility intervals of λ_{ij} and $\log_{10}(\lambda_{ij})$ are given for the j^{th} measurement on the i^{th} part for the case where ν was simulated.

Table 8 90% credibility intervals for λ_{ij} and $\log_{10}(\lambda_{ij})$ for the case where ν was simulated for the first and second measurements of the data in Table 1

Part Number	Measure-ment 1		Measure-ment 2	
	λ_{i1}	$\log_{10}(\lambda_{i1})$	λ_{i2}	$\log_{10}(\lambda_{i2})$
1	0.2104–2.9213	-0.6770–0.4656	0.1473–2.5009	-0.8309–0.3981
2	0.2140–2.9630	-0.6696–0.4717	0.2109–2.9867	-0.6759–0.4752
3	0.2047–3.0019	-0.6889–0.4774	0.2046–3.0226	-0.6891–0.4804
4	0.1998–2.9710	-0.6994–0.4729	0.2045–3.0174	-0.6893–0.4796
5	0.2041–2.9791	-0.6902–0.4741	0.2000–2.8625	-0.6990–0.4567
6	0.2091–3.0294	-0.6796–0.4814	0.1834–2.7852	-0.7366–0.4449
7	0.1704–2.6145	-0.7685–0.4174	0.1942–2.9645	-0.7118–0.4704
8	0.2034–2.9875	-0.6916–0.4753	0.2072–2.9645	-0.6836–0.4720
9	0.1987–2.8872	-0.7018–0.4605	0.2124–3.0061	-0.6728–0.4780
10	0.2053–2.9108	-0.6876–0.4640	0.2037–2.9831	-0.6910–0.4747
11	0.1782–2.7510	-0.7491–0.4395	0.2073–2.8716	-0.6834–0.4581
12	0.1786–2.5770	-0.7481–0.4111	0.2115–2.8525	-0.6747–0.4552
13	0.1248–2.1969	-0.9038–0.3418	0.1623–2.7152	-0.7897–0.4338
14	0.1984–2.9595	-0.7025–0.4712	0.1847–2.8386	-0.7335–0.4531
15	0.2047–2.9528	-0.6889–0.4702	0.1951–2.8532	-0.7097–0.4553
16	0.2095–2.9632	-0.6788–0.4718	0.2081–3.0224	-0.6817–0.4809
17	0.1985–2.9585	-0.7022–0.4711	0.1996–2.9745	-0.6998–0.4734
18	0.2024–3.0077	-0.6938–0.4782	0.1896–2.8209	-0.7222–0.4504
19	0.2079–2.9697	-0.6821–0.4727	0.1993–2.9732	-0.7005–0.4732
20	0.1934–2.8861	-0.7135–0.4603	0.2183–2.9522	-0.6609–0.4701
21	0.1584–2.5484	-0.8002–0.4063	0.1686–2.6414	-0.7731–0.4218
22	0.0337–0.8379	-1.4724– -0.0768	0.1292–2.5478	-0.8887–0.4062
23	0.1910–2.9280	-0.7190–0.4666	0.1767–2.6822	-0.7528–0.4285
24	0.1383–2.3124	-0.8592–0.3641	0.1571–2.6094	-0.8038–0.4156

By considering Table 8, it can be seen that the first measurement of part 22 can be considered an outlying observation since the upper limit of the 90% equal tails credibility interval of $\log_{10}(\lambda_{i=22,j=1})$ is below zero. This is to be expected since the first measurement of part 22 i.e. 439.5, is substantially larger than the rest of the observations in Table 1. All other 90% equal tails credibility intervals for $\log_{10}(\lambda_{ij})$ covered zero, resulting in none of the other measurements being considered as possible outliers.

9. Concluding remarks

Although the standard one-way variance components model has been widely studied, not many authors have investigated the model in cases where the measurement error model does not have the standard $N(0, \sigma_\epsilon^2)$ form. We have shown, however, that this model can be extended to the case where the measurement error model has a Student t distributional form. The Bayesian approach has several advantages for this type of problem, since it provides the flexibility to fit non-standard measurement system models and generate analyses quickly, using, Gibbs sampling in this case.

Although it was pointed out that Gibbs sampling produces dependent draws from the joint posterior density, Wilson et al. (2004) and others indicated that dependence between draws can be reduced by retaining every k^{th} draw. As mentioned for the data in Table 1, every 10^{th} draw was retained.

In this paper, the flexibility and unique features of Bayesian simulation methods, for estimating variance components and tolerance intervals, were illustrated for the mentioned one-way random effects model with non-standard measurement errors. It was also illustrated that final results are in terms of common statistics and histograms, thus providing a straightforward means of communication to investigators.

In future research, Bayesian inference in generalized error (exponential power distribution) and generalized Student t models will also be considered for these type of problems. The Bayesian chi-squared test for goodness-of-fit (Johnson, 2004) can be used for model comparisons. The calculations can be done using the Markov Chain Monte Carlo draws. A readable explanation of the Bayesian chi-squared goodness-of-fit calculations can be found in Weaver and Hamada (2008).

Based on the above information, it is hoped that this model will significantly contribute towards variance component estimation and the determination of tolerance intervals, since it is an important ingredient in the design and production of high quality, high reliability measurement systems and production processes.

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10. Appendix

Proof of Theorem 7.1.1

Function measurements are generated by the same model as defined in equation 2, i.e.

$$y_{fi} = \theta + \mu_f + \epsilon_{fj^*} \text{ for } j^* = 1, 2, \dots, q$$

$$\text{where } \epsilon_{fi} \mid \lambda_{fj^*} \sim N\left(0, \frac{\sigma_\epsilon^2}{\lambda_{fj^*}}\right).$$

From this it follows that $\bar{y}_f = \frac{1}{q} \sum_{j=1}^q y_{fj^*}$ is normally distributed with mean

$$E(\bar{y}_f | \theta, \mu_f, \sigma_\epsilon^2, \sigma_p^2, \lambda_{f1}, \dots, \lambda_{fq}) = \theta + \mu_f$$

and

$$\begin{aligned} Var(\bar{y}_f | \theta, \mu_f, \sigma_\epsilon^2, \sigma_p^2, \lambda_{f1}, \dots, \lambda_{fq}) &= \frac{1}{q^2} \sum_{j=1}^q Var(y_{fj^*}) \\ &= \frac{1}{q^2} \sum_{j^*=1}^q \frac{\sigma_\epsilon^2}{\lambda_{fj^*}} = \frac{\sigma_\epsilon^2}{q^2} \sum_{j^*=1}^q \frac{1}{\lambda_{fj^*}} \end{aligned}$$

Since $\mu_f \sim N(0, \sigma_p^2)$ it follows that

$$E(\bar{y}_f | \theta, \sigma_\epsilon^2, \sigma_p^2, \lambda_{f1}, \dots, \lambda_{fq}) = \theta$$

and

$$Var(\bar{y}_f | \theta, \sigma_\epsilon^2, \sigma_p^2, \lambda_{f1}, \dots, \lambda_{fq}) = \frac{1}{q^2} \sum_{j^*=1}^q \frac{1}{\lambda_{fj^*}} + \sigma_p^2$$

where

$$p(\lambda_{fj^*}) = \frac{\nu^{\frac{1}{2}} \nu}{2^{\frac{\nu}{2}}} \lambda_{fj^*}^{\frac{1}{2} \nu - 1} \exp\left(-\frac{1}{2} \lambda_{fj^*} \nu\right)$$

Proof of Theorem 7.1.2

Function measurements of the i^{th} part are generated by the model $\tilde{y}_{ij^*} = \theta + \mu_i + \epsilon_{ij^*}$ for $j^* = 1, \dots, q$

where

$$\epsilon_{ij^*} \sim pN\left(0, \frac{\sigma_\epsilon^2}{\lambda_{i1j^*}}\right) + (1-p)N\left(0, \frac{\sigma_\epsilon^2}{\lambda_{i2j^*}}\right) \text{ for } j^* = 1, \dots, q.$$

From this it follows that $\tilde{\bar{y}}_i = \frac{1}{q} \sum_{j^*=1}^q \tilde{y}_{ij^*}$ is normally distributed with mean

$$E(\tilde{\bar{y}}_i | \theta, \mu_i, \sigma_\epsilon^2, \sigma_p^2, \lambda_{i11}, \dots, \lambda_{i2q}) = \theta + \mu_i$$

and variance

$$\begin{aligned} & Var\left(\bar{y}_{i.} \mid \theta, \mu_i, \sigma_\epsilon^2, \sigma_p^2, \lambda_{i11}, \dots, \lambda_{i2q}\right) \\ &= \frac{\sigma_\epsilon^2}{q^2} \left\{ p \sum_{j^*=1}^q \frac{1}{\lambda_{i1j^*}} + (1-p) \sum_{j^*=1}^q \frac{1}{\lambda_{i2j^*}} \right\}. \end{aligned}$$

For our case $p = \frac{1}{2}$, therefore

$$Var\left(\bar{y}_{i.} \mid \theta, \mu_i, \sigma_\epsilon^2, \sigma_p^2, \lambda_{i11}, \dots, \lambda_{i2q}\right) = \frac{\sigma_\epsilon^2}{2q^2} \left\{ \sum_{j^*=1}^q \frac{1}{\lambda_{i1j^*}} + \sum_{j^*=1}^q \frac{1}{\lambda_{i2j^*}} \right\}.$$

Also

$$p(\lambda_{i1} \mid \mathbf{y}, \boldsymbol{\mu}, \theta, \sigma_\epsilon^2, \sigma_p^2) \propto \lambda_{i1}^{\frac{1}{2}(\nu+1)-1} \exp \left\{ -\frac{1}{2} \lambda_{i1} \left[\nu + \frac{1}{\sigma_\epsilon^2} (y_{i1} - \theta - \mu_i)^2 \right] \right\}$$

and

$$p(\lambda_{i2} \mid \mathbf{y}, \boldsymbol{\mu}, \theta, \sigma_\epsilon^2, \sigma_p^2) \propto \lambda_{i2}^{\frac{1}{2}(\nu+1)-1} \exp \left\{ -\frac{1}{2} \lambda_{i2} \left[\nu + \frac{1}{\sigma_\epsilon^2} (y_{i2} - \theta - \mu_i)^2 \right] \right\}.$$

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